

Name:

Class:

Date:

Unit 2 Vocabulary

Data · Study Sheet



Know These Cold

Every term here can show up on the AP exam. Fold the page to hide the definitions and quiz yourself, or quiz a partner.

2.1 Binary Numbers

Bit

short for binary digit; the smallest unit of data, 0 or 1

Analog data

data with values that change smoothly over time (pitch, volume, position)

Sampling

approximating analog data digitally by measuring values at discrete intervals

Overflow error

occurs when a number is too large for the fixed number of bits used to store it

Roundoff error

real numbers are approximated because only a fixed number of bits is used

2.2 Data Compression

Data compression

reducing the number of bits needed to represent data

Lossless

compression where no information is lost; the original can be restored

Lossy

compression that permanently removes some data

Run-length encoding

replaces runs of repeated values with a count and the value

Redundancy

repeated patterns in data that compression takes advantage of

2.3 Extracting Information from Data

Information

the collection of facts and patterns extracted from data

Correlation

a relationship between two variables; one changes as the other changes

Data cleaning

making data uniform without changing its meaning (e.g., fixing spellings/abbreviations)

Scalability

a system's capacity to change in size and scale to meet new demands

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Bias in data

caused by the type or source of data; NOT fixed by just collecting more data

2.4 Using Programs with Data

Filtering

choosing a subset of data that meets certain criteria

Sorting

arranging data into order (alphabetical, numerical, by date)

Data visualization

a graphical representation that reveals patterns

Transforming

changing every element in a data set in the same way (e.g., °F → °C)

Iteration

repeating steps for every element in a data set